Can error-driven discriminative learning explain transitional probability learning during listening to running speech?

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how do language users learn linguistic sequences?

relevance

sequence learning is necessary for **almost every level and modality of language**, e.g.:

- **speech comprehension** – unfolding of speech signal; from co-articulation to syntax, discourse, etc.
- **speech and language production** – incl. speech, signing, writing, typing...

and perhaps most famously, word segmentation (Saffran, Newport, & Aslin, 1996)

research question

We investigated **which learning mechanism underlies incremental learning of sequences**

- transitional probability?
- or error-driven learning?

for comparability, we used the same stimuli as Saffran et al. (1996)

Raw data and simulation visualisations

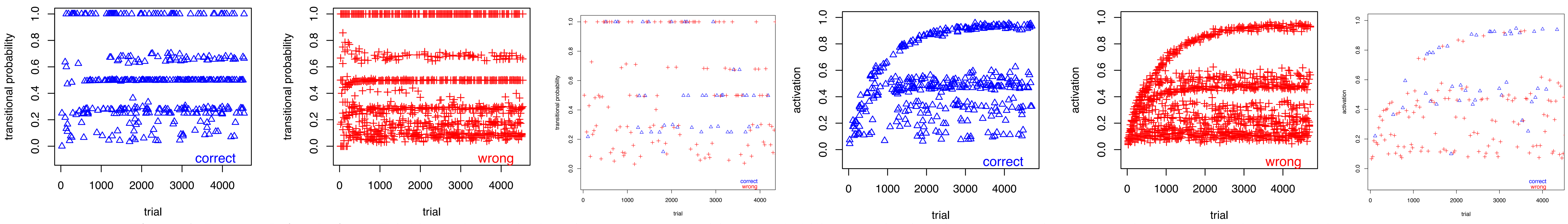
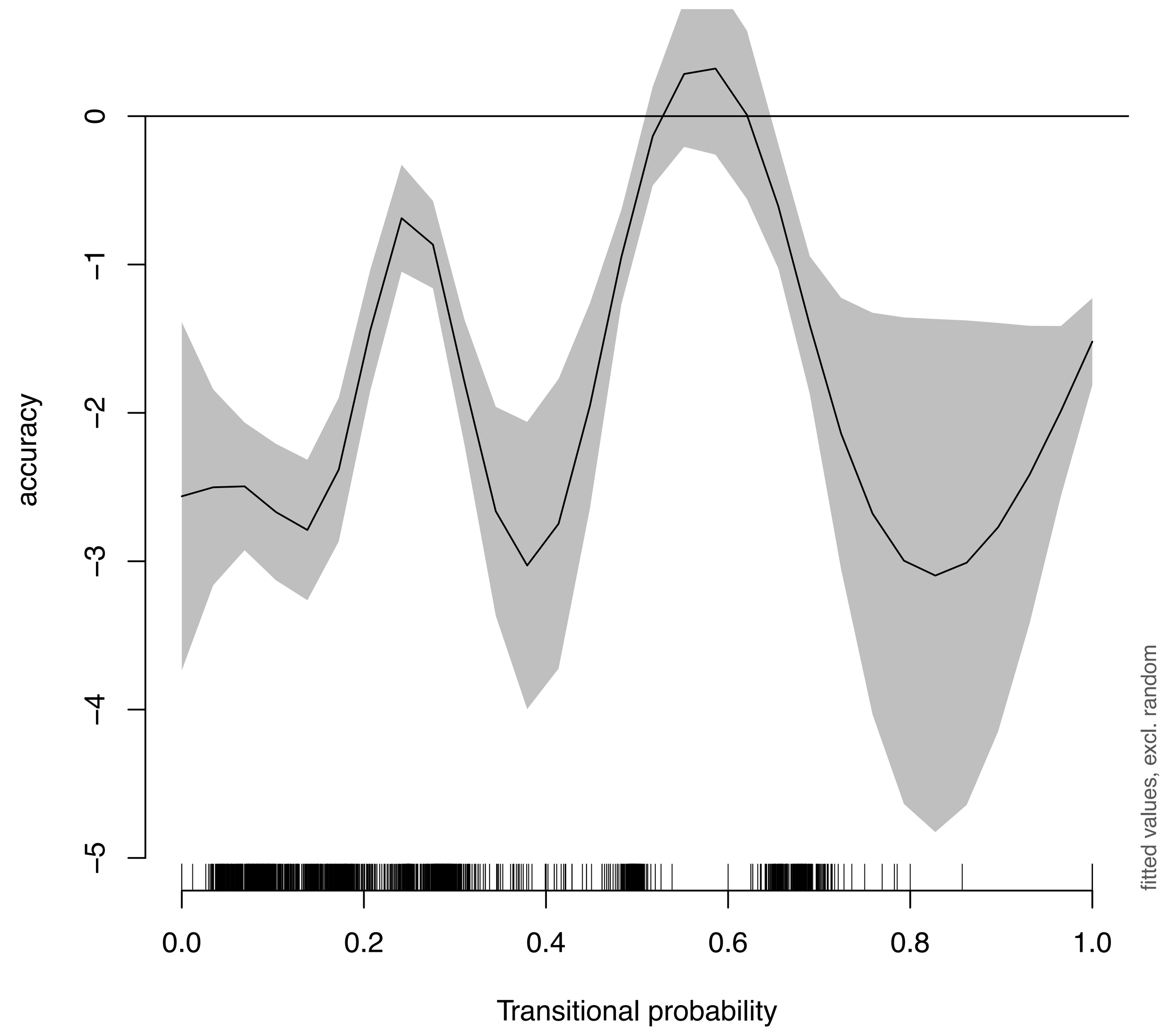
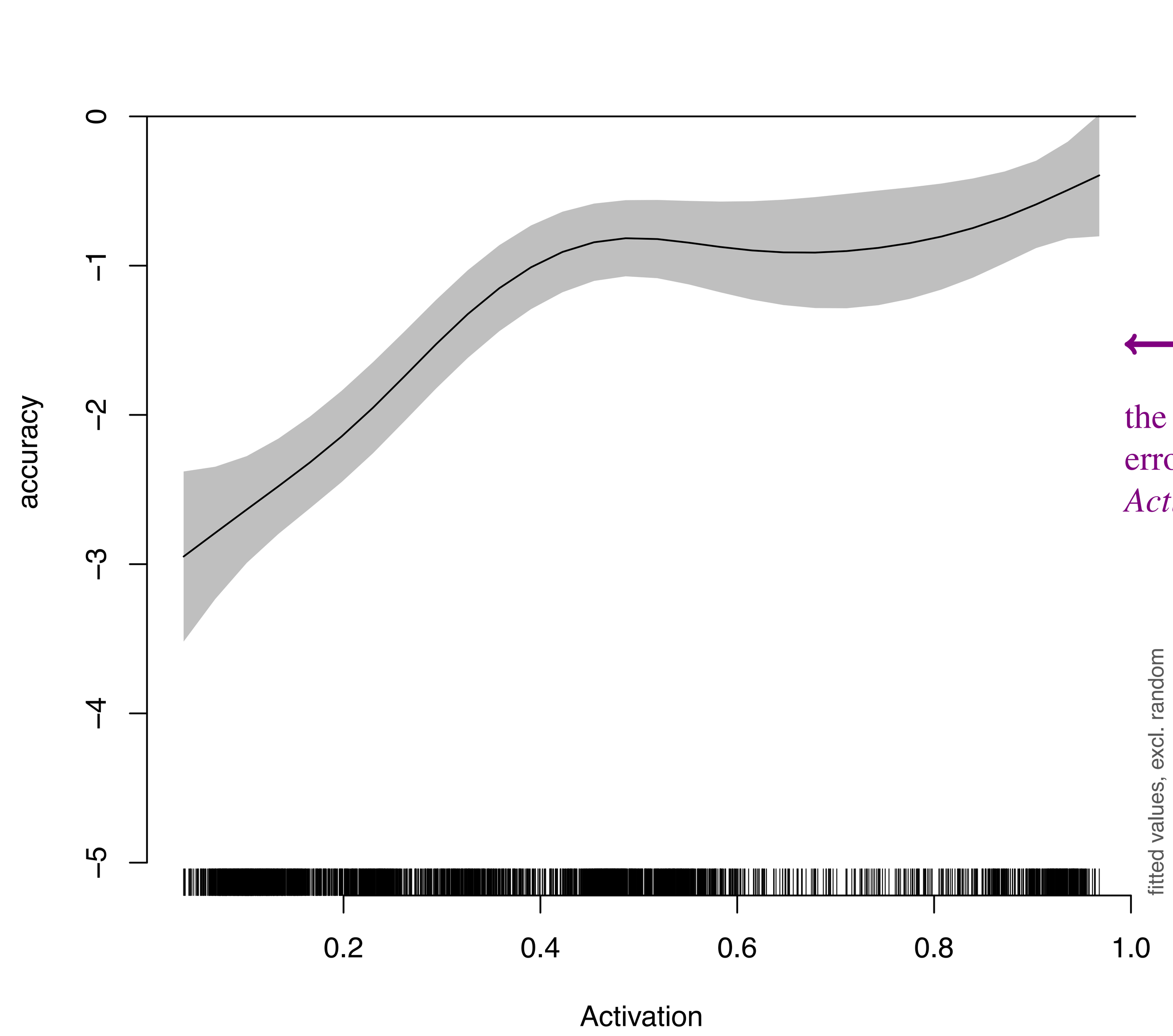


Figure 1. Data and simulations. Trial is on the x-axis; learning measure (transitional probability or activation) is on the y-axis; accuracy is colour-coded: correct (blue); wrong (red)

GAMM model estimates

transitional probability model:
there are *fluctuations* in accuracy as transitional probability increases
– this is unexpected and difficult to explain if TP drives learning





the better model:
error-driven learning
Activation

two GAMM models:
compareML()
fREML difference score 12.4

method

Based on Saffran et al. (1996), running speech stream ...

But to investigate ***incremental*** **learning and prediction**
⟶ speech stream stopped at pseudorandom intervals (marked with **[prompt]**)

babupututibubupa[prompt_1]...dababupubupadadu[prompt_2]... tabapatubipidabututibudutaba[prompt_3]...babupubupadapida [prompt_4]...bupatubitutibubabupu[prompt_5]...bupadatutibu patu[prompt_6]... bidutaba.......(and so on; 6 words, 300 x each; no prosodic cues)

16 native German participants were asked to **type the next syllable**...
Correct responses: da... ta... ba... bu... bu... bi...

simulations: two different learning measures

for *each participant* on *each trial*

transitional probability

babupubupadadutabababupututi bu bu pada...
babupubupadadutabababupututibu bu pa da...
babupubu**pada**dutabababupututibubu pa da ...

transitional probability of **da** given **pa** = $\frac{\textbf{frequency of pada}}{\textbf{frequency of pa}}$

error-driven learning: 'activation'

ba bupubupadadutaba... : #ba ba# → bu
ba bu pubupadadutababa : #bu bu# → pu
babu pu bubupadadutababa : #pu pu# → bu

Rescorla-Wagner learning equations:(1) develop expectations about next syllable from cue sequence

background

Saffran et al. showed listeners can **learn from structure in sequences**

- participants selected 'words' more often than chance in a forced-choice task after training
  - interpreted as learning from **transitional probability**

Since then, much work *assumes* sequence learning involves transitional probability

but our recent work suggests phonetic learning may instead be error-driven:

- Phonetic learning is error-driven, not just statistical (Nixon, 2020, *Cognition*)
- error-driven learning model of infant learning from acoustic signal (Nixon & Tomaschek, 2020, 2021, *Cognition*)
- Response times in sequence learning predicted by error-driven learning (Tomaschek, Ramscar, & Nixon, 2024, *Cognitive Science*)

Error-driven learning: a predictive process that discriminates between informative and uninformative cues
(Nixon, 2020; Ramscar, Dye, & McCauley, 2013)

discussion and conclusion

- Saffran et al used a 2AFC task *after training* using summed transitional probabilities
- the present study tests *incremental learning*:
  - simulated degree of expectation of each upcoming syllable *througout the experiment*
  - tested simulation predictions against by-trial accuracy

Transitional probability may be a good approximation on **summed data** (e.g. Saffran et al., 1996)

⟶ but with the increased resolution of **trial-by-trial testing** we see that **error-driven learning** seems to be the better model of incremental sequence learning